



THE EMERGENCE AND IMPACT OF NEW TECHNOLOGIES

HSC DESIGN & TECHNOLOGY

LINK TO SYLLABUS

H6.2 critically assesses the emergence and impact of new technologies, and the factors affecting their development.

- emerging technologies
 - factors affecting their development
 - criteria for evaluation
 - impact on society and the environment
 - impact on innovation.

- appraise the ecological, economic, social, ethical, and legal implications of new and emerging technologies
- analyse the impact of emerging technologies on innovation.

EMERGING TECHNOLOGIES

- WHAT DOES IT MEAN BY
“TECHNOLOGIES”?

EMERGING TECHNOLOGIES

- “TECHNOLOGIES” ARE THE **PROCESSES** THAT HUMANS USE TO INCREASE THEIR **CONTROL** OVER AND **UNDERSTANDING** OF THE **MATERIAL ENVIRONMENT** IN WHICH THEY OPERATE.
- TECHNOLOGIES COULD BE USED TO DESCRIBE **MATERIALS, TOOLS AND TECHNIQUES**.
- NEW AND EMERGING TECHNOLOGIES = **RECENTLY** / IN PROCESS OF BEING **DEVELOPED**.

MATERIAL EXAMPLES OF EMERGING TECHNOLOGIES

- **Bioplastics** are plastics derived from renewable biomass sources, such as vegetable fats and oils, corn starch, or microbiota
- Conduct a Life Cycle Analysis on bioplastics.

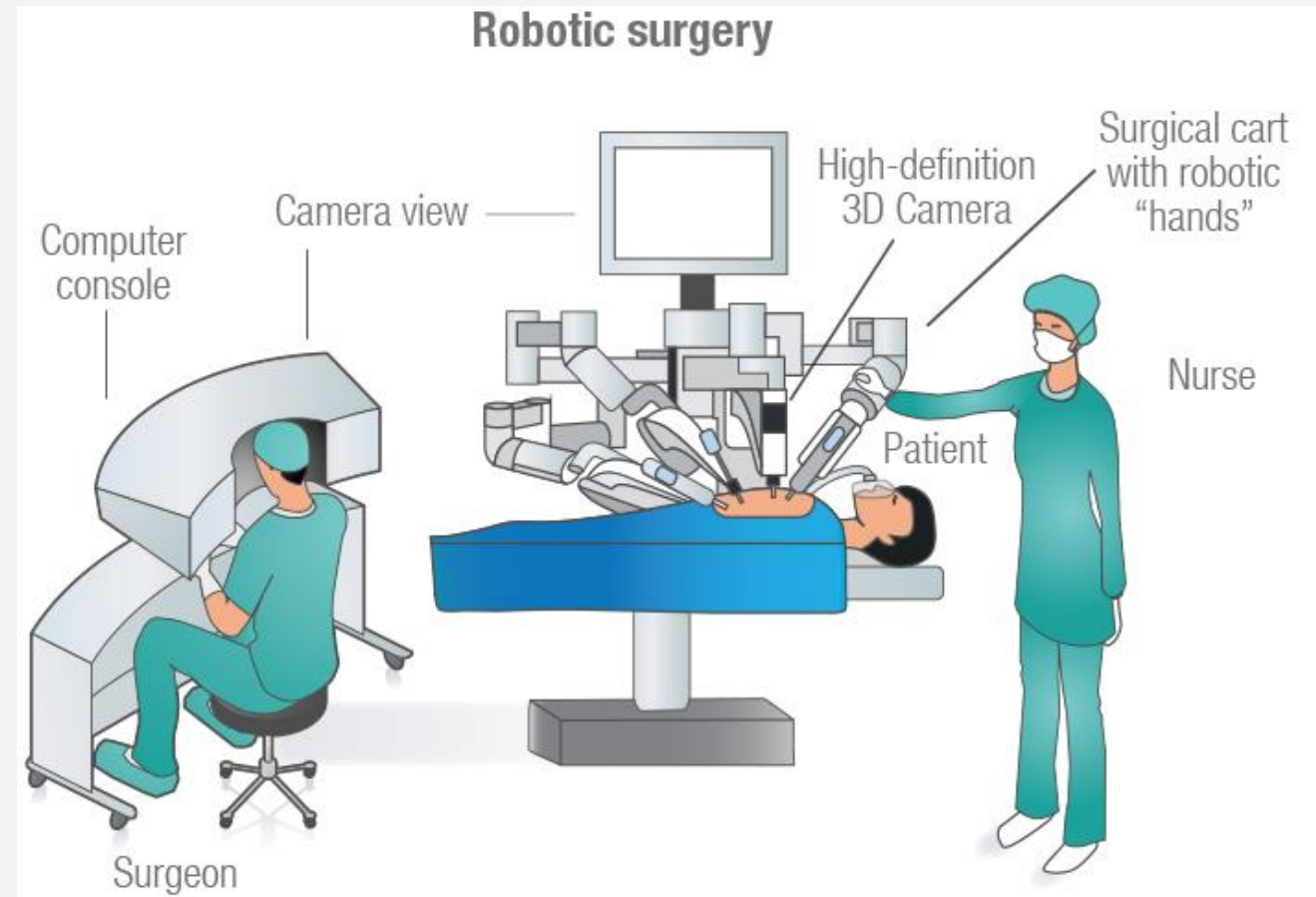


- **Programmable Matter:** Is **matter** which has the ability to change its physical properties (shape, density, moduli, conductivity, optical properties, etc.) in a **programmable** fashion, based upon user input or autonomous sensing.
- What are some of the impacts this could have on the individual and society?
- <https://www.youtube.com/watch?v=JN7BUKb0OIA>
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TOOL / TECHNIQUE

EXAMPLE OF EMERGING TECHNOLOGIES

- **Robotic surgery** an innovative tool / surgical technique that uses a robot assist surgeons to perform intricate procedures that involve small incisions that cannot be done with human hands.



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- **FACTORS AFFECTING
THE DEVELOPMENT
OF EMERGING
TECHNOLOGIES**

FACTORS AFFECTING THE DEVELOPMENT OF EMERGING TECHNOLOGIES E.G. SUSTAINABILITY

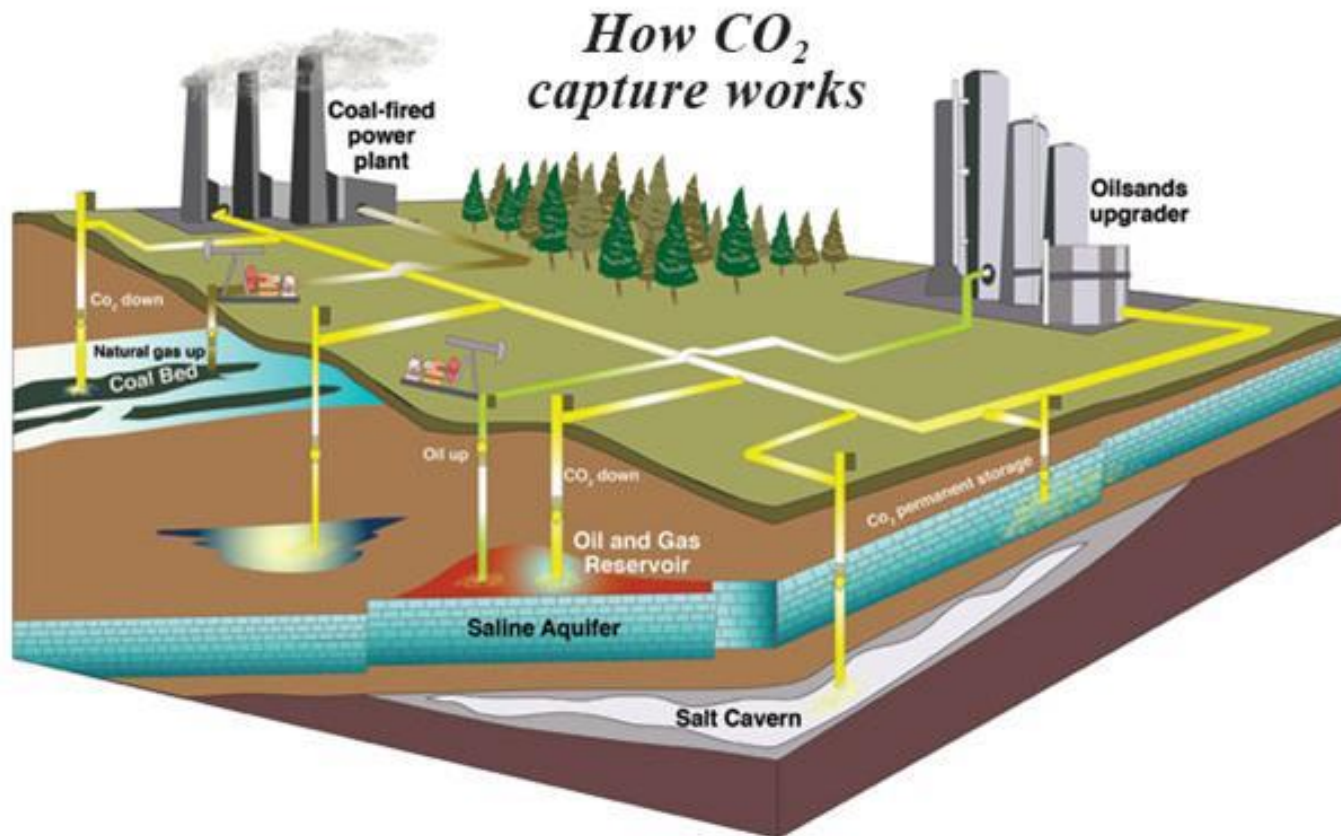
FACTOR: SUSTAINABILITY

- As we become more environmentally conscious, technologies such as solar power and wind mills are examples of emerging technologies that have been developed to conquer the preservation of our natural resources.



FACTORS AFFECTING THE DEVELOPMENT OF EMERGING TECHNOLOGIES

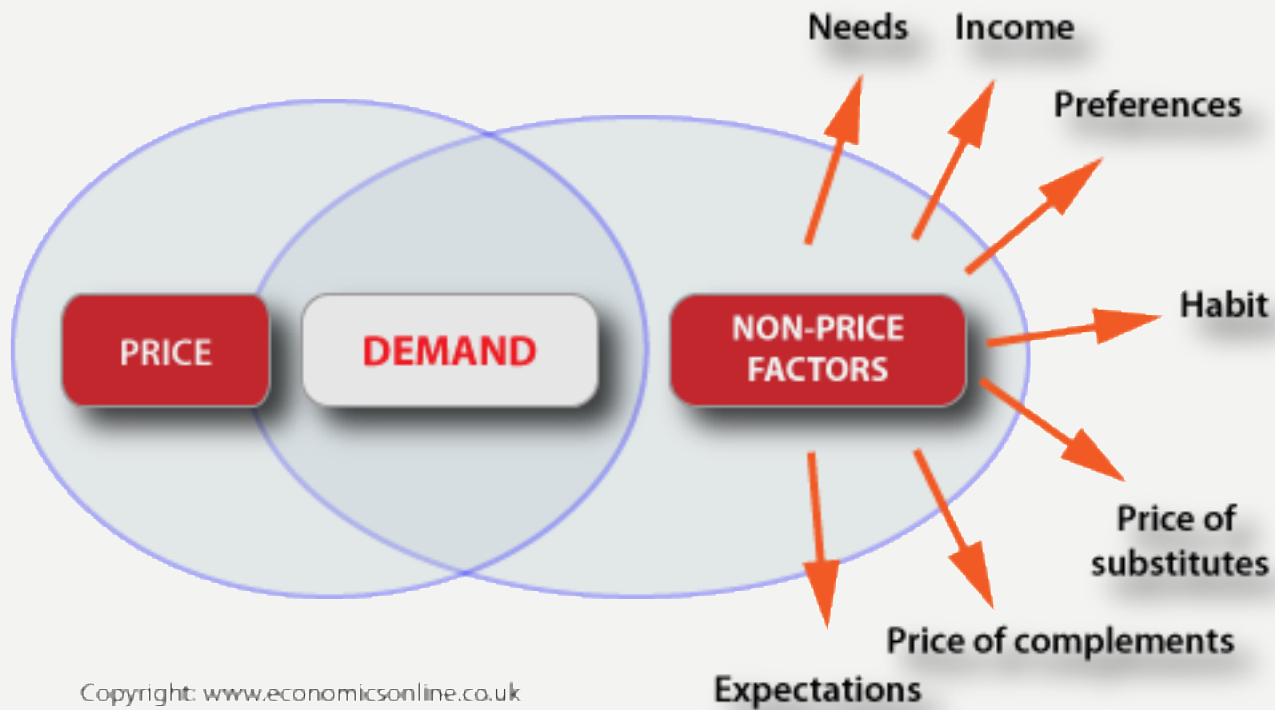
FACTOR: GLOBAL WARMING



- **Carbon capture and storage (CCS)** is the process of capturing waste CO₂ from power plants, the water or air and storing it permanently in the ground. to prevent it from going into the atmosphere.

FACTORS AFFECTING THE DEVELOPMENT OF EMERGING TECHNOLOGIES

FACTOR: CONSUMER DEMAND



- **Apple iPhone:** New technologies lead to newer technologies. Consumers expect products to be work faster, be more efficient. As markets become more competitive, companies work hard to be the “best” and win consumers over.